Test IS2023 P2 Second Partial

Test ID: 261442
Folder: /Preview
Version: 5.2
Randomised: No
Last modified: Monday, 16 december 2024 23:00:00
Number of questons: 31
Blocks: Fixed
Display questions once: No
Tools: Calculator extended
Test time: -
Maximum score: 118 pt.
Chance score: 18.51 pt. / 16%
In test set with: -

Vagueness, Uncertainty & Bayesian Networks

Question order: Fixed
The following set of questions are about uncertainty and Bayesian networks.

There are 5 questions for a total of 13 points.

Question 1 – Uncertainty versus Vagueness (elephant) – 205064.2.0

Question type: Multiple choice
Pre-test item: No
Folder: /Top/SIS/XB_0031/Old questions uncertainty and vagueness
Answer option order: Random
Partial scoring: No
Maximum score: 2
Chance score: 1.00 pt. / 50%
Status:
Last modified: 20/12/2023 16:17
Attributes: *Moeilijkheid/Difficulty* *02 Makkelijk/Easy*
Taxonomie/Taxonomy *01 Kennis/Knowledge*

Consider the statement: “All elephants have some greyish color”. Which of the following statements is true?

A The author of this statement describes the color of elephants with a degree of vagueness.

B The author of this statement describes the color of elephants with a degree of uncertainty.

Question 2 – Calculating Fuzzy Logic – 404038.2.1

Question type: Fill in (numerical)
Pre-test item: No
Folder: /Top/SIS/XB_0031/Vagueness, Uncertainty & Bayesian Networks
Answer option order: Fixed
Partial scoring: No
Maximum score: 6
Chance score: 0.00 pt. / 0%
Status:
Last modified: 14/12/2023 21:44
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

In Fuzzy Logic, the meaning of a formula is defined as a value between 0 and 1. What would be the meaning of f(((Q & P) v P) -> (¬Q v ¬P)) if f(P)=0.7 and f(Q)=0.4?

Provide a number rounded to **one** digit after the decimal point. Use dot as a separator (e.g. 0.22).

0.9

Question 3 – Fuzzy Relations – 405474.3.1

Question type: Fill in (multiple)
Pre-test item: No
Folder: /Top/SIS/XB_0031/Fuzzy Relations
Answer option order: Fixed
Partial scoring: Yes
Maximum score: 6
Chance score: 0.00 pt. / 0%
Status:
Last modified: 19/12/2023 15:38
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

Consider the following relations, where WGS-5 and WGS-6 stands for Working Group Session 5 and Working Group Session 6, respectively.

Understanding:

	WGS5	WGS6
Joana	0.3	0.7
Andreea	0.6	0.4
Francesca	0.8	0.2

Contains;

	Fuzzy Logic	Probability
WGS-5	0.7	0.3
WGS-6	0.4	0.6

Fill in the table for the composition of the two relations.

	Fuzzy Logic	**Probability**
Joana	0.4	0.6
Andreea	0.6	0.4
Francesca	0.7	0.3

[Alphanumeric]	[Alphanumeric]	[Alphanumeric]	[Alphanumeric]	[Alphanumeric]
0.4	**0.6**	**0.6**	**0.4**	**0.7**
0,4	0,6	0,6	0,4	0,7

[Alphanumeric]
0.3
0,3

Question 4 – Joint Probability Distribution – 404033.3.0

Question type: Multiple choice
Pre-test item: No
Folder: /Top/SIS/XB_0031/Vagueness, Uncertainty & Bayesian Networks
Answer option order: Random
Partial scoring: No
Maximum score: 5
Chance score: 1.25 pt. / 25%
Status:
Last modified: 20/12/2023 16:58
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

Consider the following joint probability distribution table.

	headache		**¬ headache**	
	fever	**¬ fever**	**fever**	**¬ fever**
cold	0.258	0.094	0.135	0.015
¬ cold	0.012	0.063	0.036	0.387

What is the probability of **P**(Fever | headache)?

A < 0.63, 0.37 >

B < 0.37, 0.63 >

C < 0.30, 0.70 >

D < 0.78, 0.22 >

Question 5 – Independent vs. Dependent Events – 404034.2.1

Question type: Multiple choice
Pre-test item: No
Folder: /Top/SIS/XB_0031/Vagueness, Uncertainty & Bayesian Networks
Answer option order: Random
Partial scoring: No
Maximum score: 3
Chance score: 0.75 pt. / 25%
Status:
Last modified: 20/12/2023 18:21
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

Two fair dice are thrown, and the numbers uppermost are observed. Decide whether the following events are dependent or independent and what are their probabilities:

A: The difference between the numbers on the two dice is 3.
B: The number on the red die is odd.

A Dependent, P(B) = 12/36, P(B|A) = 1/3

B Independent, P(A) = 6/36, P(A|B) = 1/6

C Dependent, P(A) = 1/6 , P(A|B) = 3/9

D Independent, P(B) = 6/36, P(B|A) = 1/6

Question 6 – Bayes Rule – 404035.4.0

Question type: Fill in (multiple)
Pre-test item: No
Folder: /Top/SIS/XB_0031/Vagueness, Uncertainty & Bayesian Networks
Answer option order: Fixed
Partial scoring: Yes
Maximum score: 8
Chance score: 0.00 pt. / 0%
Status:
Last modified: 20/12/2023 16:58
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

Given the following information:

- 219 out of 365 days in Amsterdam are rainy.
- Four out of five days, Anne travels with public transport when it is raining.
- Anne travels with public transport 64% of the time regardless of the weather.

For the following questions, provide a number rounded to **two** digits after the decimal point. Use dot as a separator (e.g. 0.22).

What is the probability of a rainy day in Amsterdam? 0.60

What is the probability that Anne is traveling with public transport given that it is raining? 0.80

What is the probability that Anne is traveling with public transport regardless of the weather? 0.64

What is the probability that today is a rainy day given that Anne has traveled with public transport? 0.75

[Numeric]	[Numeric]	[Numeric]	[Numeric]
0.60	**0.80**	**0.64**	**0.75**

Question 7 – Bayesian network (advanced) – 292213.3.1

Question type: Fill in (multiple)
Pre-test item: No
Folder: /Top/SIS/XB_0031/New questions - final exam 2021/2022
Answer option order: Fixed
Partial scoring: No
Maximum score: 5
Chance score: 0.00 pt. / 0%
Status:
Last modified: 21/02/2024 11:46
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

Given the following Bayesian network, please fill in the blank:

B	P(B)
+b	0.001
-b	0.999

E	P(E)
+e	0.002
-e	0.998

B	E	A	P(A|B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

A	J	P(J|A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

A	M	P(M|A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

This question requires some advanced calculation. Make sure you do not loose too much time on it.

What is the probability that the alarm has been activated, but neither burglary nor earthquake has occurred, and both John and Mary call?

(Hint: The full joint distribution is the product of the local conditional distributions P(J ∧ M ∧ A ∧ ¬B ∧ ¬E)) 0.000628

[Numeric]

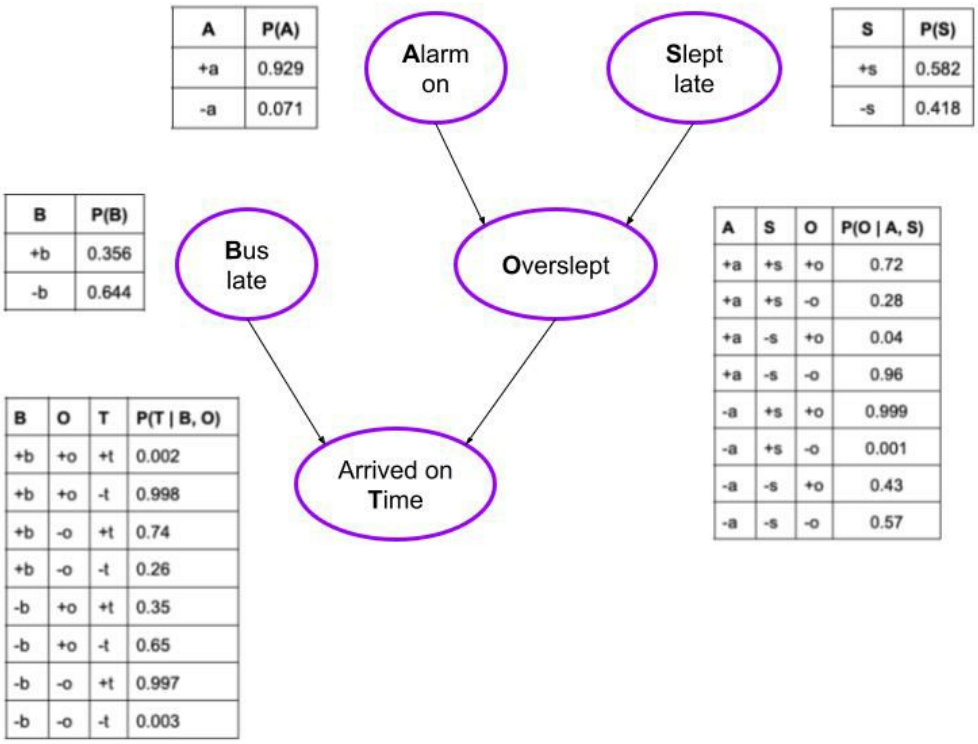
0.000628

0.0006280

Question 8 – Bayesian Networks (Calculations 2) – 404043.3.3

Question type: Fill in (multiple)
Pre-test item: No
Folder: /Top/SIS/XB_0031/Vagueness, Uncertainty & Bayesian Networks
Answer option order: Fixed
Partial scoring: No
Maximum score: 7
Chance score: 0.00 pt. / 0%
Status:
Last modified: 16/12/2024 23:01
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

Consider the following Bayesian network and the accompanying probability tables about the high school student Tom.



A	P(A)
+a	0.929
-a	0.071

S	P(S)
+s	0.582
-s	0.418

B	P(B)
+b	0.356
-b	0.644

A	S	O	P(O | A, S)
+a	+s	+o	0.72
+a	+s	-o	0.28
+a	-s	+o	0.04
+a	-s	-o	0.96
-a	+s	+o	0.999
-a	+s	-o	0.001
-a	-s	+o	0.43
-a	-s	-o	0.57

B	O	T	P(T | B, O)
+b	+o	+t	0.002
+b	+o	-t	0.998
+b	-o	+t	0.74
+b	-o	-t	0.26
-b	+o	+t	0.35
-b	+o	-t	0.65
-b	-o	+t	0.997
-b	-o	-t	0.003

Provide numbers rounded to **three** digits after the decimal point. Only round at the very end of your calculations. Use dot as a separator (e.g. 0.123).

What is the probability that Tom arrives on time for his classes? 0.594

[Numeric]

0.594

Machine Learning

Question order: Fixed
The following set of questions are about the Machine Learning 1 lecture.

There are 4 questions for a total of 9 points.

Question 9 – Regression, Classification, Ranking, and Clustering – 404400.1.0

Question type: Multiple choice
Pre-test item: No
Folder: /Top/SIS/XB_0031/Machine Learning 1
Answer option order: Random
Partial scoring: No
Maximum score: 2
Chance score: 0.50 pt. / 25%
Status:
Last modified: 15/12/2023 16:52
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

What is the main difference between regression and classification in machine learning?

A Regression predicts continuous values, while classification predicts categories.

B Classification predicts continuous values, while regression predicts categories.

C Both regression and classification predict categories.

D Both regression and classification predict continuous values.

Question 10 – Cross–Validation and Occam's Razor – 404397.3.0

Question type: Multiple choice
Pre-test item: No
Folder: /Top/SIS/XB_0031/Machine Learning 3
Answer option order: Random
Partial scoring: No
Maximum score: 2
Chance score: 0.50 pt. / 25%
Status:
Last modified: 18/12/2023 12:35
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

What is the primary purpose of cross-validation in machine learning?

A Testing the model on an unseen dataset

B Training the model on multiple subsets of the dataset

C Fine-tuning hyperparameters using the test set

D Getting a reliable estimate of how well the model generalizes to unseen data

Question 11 – Types of ML Algorithms – 404048.1.0

Question type: Multiple choice
Pre-test item: No
Folder: /Top/SIS/XB_0031/Machine Learning 1
Answer option order: Random
Partial scoring: No
Maximum score: 2
Chance score: 0.40 pt. / 20%
Status:
Last modified: 14/12/2023 21:55
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

The assistant of a famous photographer has been tasked with ordering the pictures taken by the photograph in different folders based on similarities. However, there is an enormous amount of pictures, so the assistant wishes to take a short-cut instead of sorting the pictures by hand.

What type of machine learning algorithm is most appropriate for this problem?

A Classification

B Regression

C Collaborative filtering

D Ranking

E Clustering

Question 12 – ML Types: Supervised, Unsupervised – 291569.4.0

Question type: Fill in (multiple)
Pre-test item: No
Folder: /Top/SIS/XB_0031/New questions - final exam 2021/2022
Answer option order: Fixed
Partial scoring: Yes
Maximum score: 6
Chance score: 0.00 pt. / 0%
Status:
Last modified: 20/12/2023 17:01
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

What kind of model do we need for the following cases (**classification, regression, ranking, collaborative filtering, clustering**)?
First, indicate whether this would be **supervised or unsupervised** machine learning.

Example:
a) Searching for overlaps in the parts of the population affected by diabetes and arthritis.

unsupervised

clustering

a) Deciding if an MRI image shows a cancerous tumour or not.
supervised
classification

b) Twitter is a social networking site that allows micro-blogging services where people broadcast short, public messages refer as tweets. Twitter wants to find out about similar users.
unsupervised
clustering

c) Model to forecast the sales number based on the history of the sales.
Supervised
regression

d) Auto-categorizing the news provided by the News channel.
unsupervised
clustering

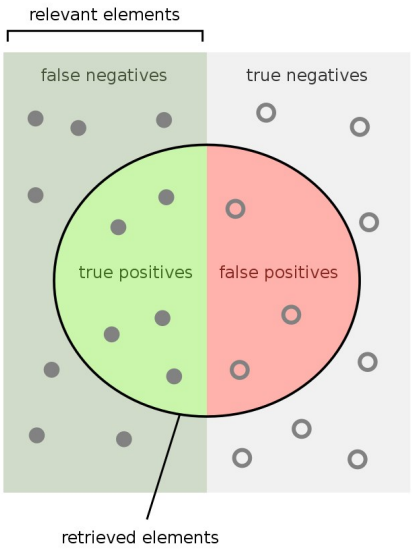
e) Suggesting a meal in the cafeteria self-order counter based on ratings of meals given by patients and hospital workers.
Supervised
collaborative

[Alphanumeric]	[Alphanumeric]	[Alphanumeric]	[Alphanumeric]	[Alphanumeric]
supervised	**classification**	**unsupervised**	**clustering**	**Supervised**
Supervised	Classification	Unsupervised	Clustering	supervised
[Alphanumeric]	[Alphanumeric]	[Alphanumeric]	[Alphanumeric]	[Alphanumeric]
regression	**unsupervised**	**clustering**	**Supervised**	**collaborative**
Regression	Unsupervised	Clustering	supervised	Collaborative

Question 13 – Precision vs Recall – 404129.3.0

Question type: Multiple response (limited)
Pre-test item: No
Folder: /Top/SIS/XB_0031/Machine Learning 2
Answer option order: Random
Partial scoring: Yes
Maximum score: 2
Chance score: 1.00 pt. / 50%
Status:
Last modified: 20/12/2023 18:47
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

Consider the following diagram and choose the right definitions.



A Precision refers to how many retrieved items are relevant.

B Precision refers to how many relevant items are retrieved.

C Recall refers to how many retrieved items are relevant.

D Recall refers to how many relevant items are retrieved.

Question 14 – Precision & Recall – 404053.2.0

Question type: Fill in (multiple)
Pre-test item: No
Folder: /Top/SIS/XB_0031/Machine Learning 2
Answer option order: Fixed
Partial scoring: Yes
Maximum score: 4
Chance score: 0.00 pt. / 0%
Status:
Last modified: 19/12/2023 15:28
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

Calculate precision and recall for the following system, which classifies pictures of insects as butterflies and NOT butterflies.

Picture	Predicted Class	Actual Class
	Butterfly	Butterfly
	NOT Butterfly	Butterfly
	Butterfly	NOT Butterfly
	Butterfly	Butterfly
	Butterfly	Butterfly
	NOT Butterfly	NOT Butterfly
	NOT Butterfly	Butterfly

Provide numbers rounded to **two** digits after the decimal point. Use dot as a separator (e.g. 0.22).

Precision: 0.75
Recall: 0.60

[Numeric]	[Numeric]
0.75	**0.60**

Question 15 – Ranking Evaluation – 404058.2.0

Question type: Multiple choice
Pre-test item: No
Folder: /Top/SIS/XB_0031/Machine Learning 2
Answer option order: Random
Partial scoring: No
Maximum score: 2
Chance score: 0.50 pt. / 25%
Status:
Last modified: 14/12/2023 22:32
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

Consider the following output of an Ranking algorithm created to retrieve movies with Keira Knightley:

1. Pirates of the Caribbean: Dead Man's Chest*
2. Barbie
3. The Imitation Game*
4. Pirates of the Caribbean: At World's End*
5. Doctor Strange
6. Pride & Prejudice*
7. Dune
8. Official Secrets*
9. Atonement*
10. A Dangerous Method*
11. The Duchess*
12. Spider-Man: Homecoming

Only the titles with a star (*) at the end represent movies in which Keira Knightley has **actually** starred.

What is the precision at 9 and the precision at 11 of this ranking?

A 8/11, 6/9

B 6/9, 8/11

C 6/9, 7/11

D 8/9, 8/11

Question 16 – Characteristics of Naive Bayes Classification – 404403.1.0

Question type: Multiple choice
Pre-test item: No
Folder: /Top/SIS/XB_0031/Machine Learning 1
Answer option order: Random
Partial scoring: No
Maximum score: 2
Chance score: 0.50 pt. / 25%
Status:
Last modified: 15/12/2023 17:34
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

What is a characteristic of Naive Bayes classifiers?

A They are sensitive to irrelevant features.

B They assume independence between features.

C They are prone to overfitting.

D They are primarily used for regression tasks.

Question 17 – Naive Bayes Classifier – 404050.2.1

Question type: Fill in (multiple)
Pre-test item: No
Folder: /Top/SIS/XB_0031/Machine Learning 1
Answer option order: Fixed
Partial scoring: Yes
Maximum score: 7
Chance score: 1.17 pt. / 17%
Status:
Last modified: 22/12/2023 12:39
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

Consider the following list of feature vectors, obtained from five documents about butterflies (B) and about moths (M).

	wings	**head**	**threadlike**	**legs**	**camouflage**	**antennae**	CLASS
1	0	0	0	1	1	1	B
2	1	0	1	0	1	0	M
3	0	1	0	1	0	1	B
4	1	0	0	0	0	0	B
5	0	1	0	0	1	1	M

Using the Boolean features **{wings, head, threadlike, legs, camouflage, antennae}**, make a Naive Bayes Classifier to classify the following document as belonging to either the Butterfly or Moth class.

		CLASS
6	The antennae on their head are composed of many segments and have threadlike tips**.**	?

What is the probability that the new document is from class M? Provide a number rounded to **five** digits after the decimal point. Use dot as a separator (e.g. 0.22). 0.00025

What is the probability that the new document is from class B? Provide a number rounded to **six** digits after the decimal point. Use dot as a separator (e.g. 0.22). 0.000198

Based on the above probabilities, therefore, the document should be classified as: Class M

[Numeric]	[Numeric]	[List]
0.00025	**0.000198**	**Class M**
		Class B

Question 18 – Feature vector extraction – 404584.12.6

Question type: Fill in (multiple)
Pre-test item: No
Folder: /Top/SIS/XB_0031/Machine Learning 2
Answer option order: Fixed
Partial scoring: Yes
Maximum score: 5
Chance score: 0.00 pt. / 0%
Status:
Last modified: 20/12/2023 18:43
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

Convert each of these sentences into feature vector after removing punctuations, stemming and stop-words (CHECK: **only verbs, adjectives and nouns are NOT stopwords in this example)**.

a. Dog walks in the field.
b. The cow sleeps on the poppies.
c. The cats and the dogs run in the field.
d. The cow is mad because the cats and the dogs walked all over the poppies.

Order the features in the order in which they appear in the sentences from left to right. The first row is for the words.

dog	walk	field	cow	sleep	poppy	cat	run	mad
1	1	1	0	0	0	0	0	0
0	0	0	1	1	1	0	0	0
1	0	1	0	0	0	1	1	0
1	1	0	1	0	1	1	0	1

[Alphanumeric]	[Alphanumeric]	[Alphanumeric]	[Alphanumeric]	[Alphanumeric]	
dog	**walk**	**field**	**cow**	**sleep**	
Dog	Walk	Field	Cow	Sleep	
[Alphanumeric]	[Alphanumeric]	[Alphanumeric]	[Alphanumeric]	[Numeric]	[Numeric]
poppy	**cat**	**run**	**mad**	**1**	**1**
Poppy	Cat	Run	Mad		

[Numeric]	[Numeric]	[Numeric]	[Numeric]	[Numeric]	[Numeric]
1	**0**	**0**	**0**	**0**	**0**
[Numeric]	[Numeric]	[Numeric]	[Numeric]	[Numeric]	[Numeric]
0	**0**	**0**	**0**	**1**	**1**
[Numeric]	[Numeric]	[Numeric]	[Numeric]	[Numeric]	[Numeric]
1	**0**	**0**	**0**	**1**	**0**
[Numeric]	[Numeric]	[Numeric]	[Numeric]	[Numeric]	[Numeric]
1	**0**	**0**	**0**	**1**	**1**
[Numeric]	[Numeric]	[Numeric]	[Numeric]	[Numeric]	[Numeric]
0	**1**	**1**	**0**	**1**	**0**
[Numeric]	[Numeric]	[Numeric]	[Numeric]		
1	**1**	**0**	**1**		

Machine Learning 2

Question order: Fixed
The following set of questions are about the Machine Learning 2 lecture.

There are 5 questions for a total of 13 points.

Question 19 – k–Means Clustering – 404404.1.0

Question type: Multiple choice
Pre-test item: No
Folder: /Top/SIS/XB_0031/Machine Learning 1
Answer option order: Random
Partial scoring: No
Maximum score: 2
Chance score: 0.50 pt. / 25%
Status:
Last modified: 15/12/2023 17:39
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

How does k-means clustering work?

A It assigns each data point to the nearest centroid.

B It calculates the mean of each feature in the dataset.

C It uses a probability distribution to cluster data points.

D It performs a linear transformation on the input features.

Question 20 – Euclidean Distance – 404130.1.2

Question type: Multiple choice
Pre-test item: No
Folder: /Top/SIS/XB_0031/Machine Learning 2
Answer option order: Random
Partial scoring: No
Maximum score: 2
Chance score: 0.50 pt. / 25%
Status:
Last modified: 04/12/2024 16:09
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

Which of the following will be Euclidean Distance between the two data points (4,6) and (6,4)?

A √4

B -4

C 2

D √8

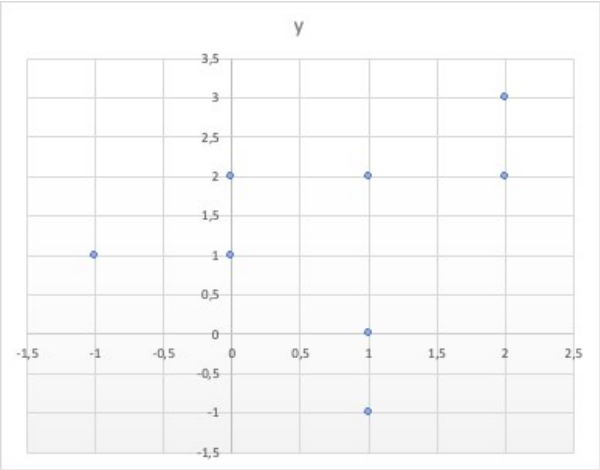
Question 21 – K Nearest Neighbours – 404131.1.1

Question type: Multiple choice
Pre-test item: No
Folder: /Top/SIS/XB_0031/Machine Learning 2
Answer option order: Random
Partial scoring: No
Maximum score: 3
Chance score: 1.00 pt. / 33%
Status:
Last modified: 20/12/2023 16:36
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

Consider the following data where x and y are 2 input variables and Class is a dependent variable.

x	y	Class
-1	1	-
0	1	+
0	2	-
1	-1	-
1	0	+
1	2	+
2	2	-
2	3	+

Below is the scatter plot of the above data.



Q. What would be the class of data point (1.5, 2.5) be using euclidean distance where k = 3 and k-nearest neighbour?

A +Class

B -Class

C None of the given options

Question 22 – Feature based Similarity – 404483.1.1

Question type: Match (even)
Pre-test item: No
Folder: /Top/SIS/XB_0031/Machine Learning 2
Answer option order: Random
Partial scoring: Yes
Maximum score: 3
Chance score: 1.00 pt. / 33%
Status:
Last modified: 17/12/2023 18:13
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

Two texts are similar when their feature vectors are similar. Which of the following corresponds to which?

Hamming Distance	(number of different bit)/(total number of bits)
Euclidean Distance	$\sqrt{\sum_{i=1}^{n}(q_i - p_i)^2}$
Cosine Similarity	$\frac{\mathbf{A}\cdot\mathbf{B}}{\|\mathbf{A}\|\|\mathbf{B}\|} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2}\sqrt{\sum_{i=1}^{n} B_i^2}},$

Question 23 – Cosine Similarity – 404057.3.0

Question type: Multiple choice
Pre-test item: No
Folder: /Top/SIS/XB_0031/Machine Learning 2
Answer option order: Random
Partial scoring: No
Maximum score: 5
Chance score: 1.25 pt. / 25%
Status:
Last modified: 19/12/2023 15:48
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

Consider the following list of feature vectors.

A	1	0	1	0	1	1	0	0
B	0	0	0	1	0	0	1	1
C	1	1	1	1	0	0	1	0
D	0	0	1	0	1	1	1	0

Which vector represents the nearest (most similar) neighbour of the below vector if we apply Cosine Similarity?

X	0	1	0	0	0	1	1	0

A A

B B

C C

D D

Machine Learning 3

Question order: Fixed
The following set of questions are about the Machine Learning
There are 4 questions for a total of 7 points.

Question 24 – Feature Engineering and Design Matrix – 404398.1.1

Question type: Multiple choice
Pre-test item: No
Folder: /Top/SIS/XB_0031/Machine Learning 3
Answer option order: Random
Partial scoring: No
Maximum score: 2
Chance score: 0.50 pt. / 25%
Status:
Last modified: 20/12/2023 16:41
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

In the context of feature creation, what is a design matrix?

A A matrix used for clustering analysis

B A matrix of new features generated from existing ones.

C A matrix representing the confusion matrix

D A matrix for storing hyperparameter values

Question 25 – Methodology and Overfitting – 404393.3.0

Question type: Multiple choice
Pre-test item: No
Folder: /Top/SIS/XB_0031/Machine Learning 3
Answer option order: Random
Partial scoring: No
Maximum score: 3
Chance score: 0.75 pt. / 25%
Status:
Last modified: 20/12/2023 18:47
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

Which of the following is a common practice in machine learning to prevent overfitting?

A Using the test set for hyperparameter tuning

B Ignoring the validation set during model training

C Using as few parameters as possible

D Increasing the complexity of the model without constraints

Question 26 – Model Fitting – 404481.3.0

Question type: Multiple choice
Pre-test item: No
Folder: /Top/SIS/XB_0031/Machine Learning 4
Answer option order: Random
Partial scoring: No
Maximum score: 5
Chance score: 1.25 pt. / 25%
Status:
Last modified: 20/12/2023 18:48
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

Consider the following simple linear regression model with the hypothesis h(x)= b*x and three data points:

Data Point 1: (2,5)
Data Point 2: (3,7) and
Data Point 3: (5,11)

Find the value of b.

A 2.263

B 3.363

C 3.636

D 2

Question 27 – Regression – 292316.3.0

Question type: Multiple choice
Pre-test item: No
Folder: /Top/SIS/XB_0031/New questions - final exam 2021/2022
Answer option order: Random
Partial scoring: No
Maximum score: 6
Chance score: 1.00 pt. / 17%
Status:
Last modified: 21/02/2024 11:25
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

This exercise is about Regression.

Let us try to fit a linear function to a number of training sets where the slope s is known to be 1.

Suppose we have three pairs of values (1,0), (2,0) and (3,1) for two features. What is the optimal fitting function in this model class (so minimises the error rate)?

A f(x)= x - 5/3

B f(x)= x - 3/5

C f(x)= x + 5/3

D f(x)= x + 3/5

E There is no optimal fitting function

F Neither of the functions in the other options is the optimal fitting function, but there exist one.

Machine Learning 4

Question order: Fixed
The following set of questions are about the Machine Learning 4 lecture.

There are 3 questions for a total of 10 points.

Question 28 – Purpose of Activation Function – 404405.1.0

Question type: Multiple choice
Pre-test item: No
Folder: /Top/SIS/XB_0031/Machine Learning 4
Answer option order: Random
Partial scoring: No
Maximum score: 2
Chance score: 0.50 pt. / 25%
Status:
Last modified: 15/12/2023 17:44
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

What is the purpose of the activation function in a neural network?

A To compute the mean of the input features

B To introduce non-linearity to the network

C To determine the learning rate during training

D To calculate the gradient of the loss function

Question 29 – Sigmoid vs Softmax function – 404128.1.0

Question type: Multiple response
Pre-test item: No
Folder: /Top/SIS/XB_0031/Machine Learning 4
Answer option order: Random
Partial scoring: Yes
Maximum score: 2
Chance score: 0.67 pt. / 33%
Status:
Last modified: 15/12/2023 10:55
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

Sigmoid and Softmax are both activation functions used in neural networks. Which of the following is true about these functions.

A Sigmoid function is used for binary classification

B Softmax function is used for binary classification

C Sigmoid function is used for multi-class classification

D Softmax function is used for multi-class classification

Question 30 – Model Finding and Optimization – 404402.2.0

Question type: Multiple choice
Pre-test item: No
Folder: /Top/SIS/XB_0031/Machine Learning 4
Answer option order: Random
Partial scoring: No
Maximum score: 2
Chance score: 0.50 pt. / 25%
Status:
Last modified: 15/12/2023 17:13
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

Which optimization technique involves adjusting model parameters to minimize a predefined loss function?

A Analytical optimization

B Uninformed optimization

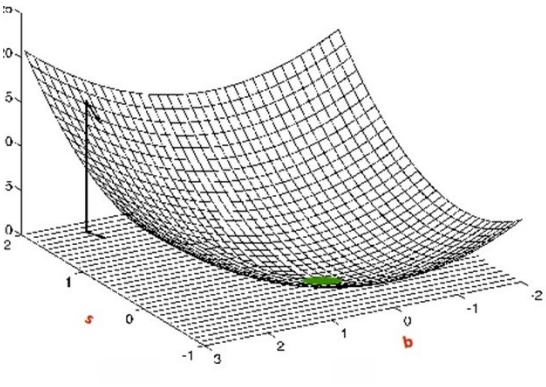
C Hyperparameter tuning

D Gradient-based optimization

Question 31 – Model space MC – 205100.3.0

Question type: Multiple response
Pre-test item: No
Folder: /Top/SIS/XB_0031/New questions - final exam 2020/2021/Questions machine learning
Answer option order: Random
Partial scoring: Yes
Maximum score: 5
Chance score: 1.53 pt. / 31%
Status:
Last modified: 20/12/2023 16:56
Attributes: *Taxonomie/Taxonomy 00 Onbekend/Unknown*

In the following image, also shown in the lecture, the green point is identified as the optimal solution to a learning problem.



Which of the following statements are true? (multiple answers are possible).

A The green point refers refers to the linear function with the lowest error rate (the one that fits a particular learning model best according to the training data)

B The green point refers to the point in the binary vectorspace with the best training example (the one that fits a particular learning model best)

C The third dimension (orthogonal to s and b) shows the error rate for a linear separator function, eg. f(x) = sx+b.

D The green point is the point where f(x) = sx+b reaches a minimum, ie where the linear separator reaches the lowest point (in other words, f(x)=0

E The third dimension (orthogonal to s and b) enumerates all possible linear separators, so that the green point is the optimal one (as it is the lowest).

F There are two ways to find the best machine learning model in this scenario: either we analytically find the green point as the point where the derivation of the error function is 0, or we apply gradient descent as a search method to move step by step to the green point. (Gradient descent is a local search similar to hill-climbing)

G With a Neural Network we can predict the value of the 3rd dimension (orthogonal to s and b on the x and y axis) given the representation of the features according to variables s and b.